



Star-Galaxy Separation in Fornax

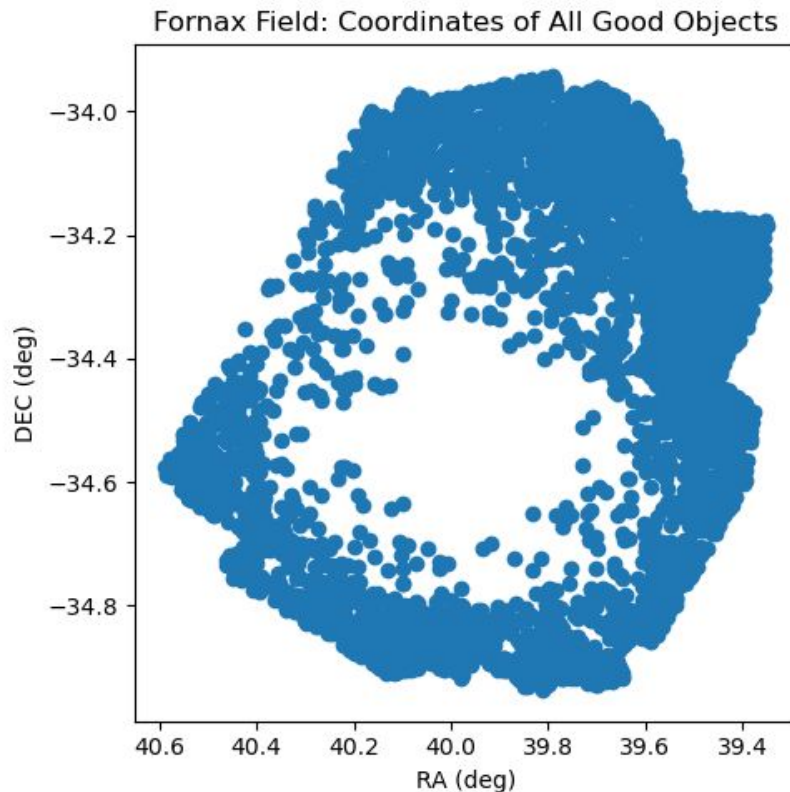
Sprint Day: Checking for RGB Contamination

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Created Fornax Field Star and Galaxy Catalogs

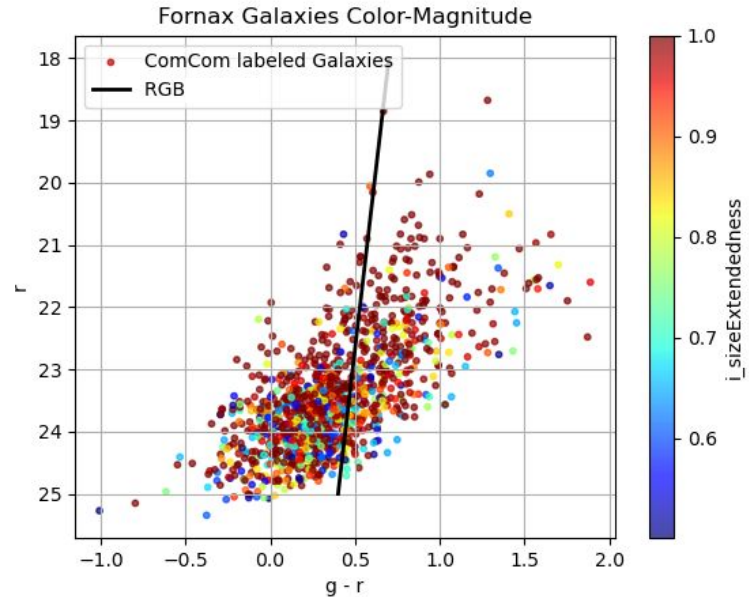
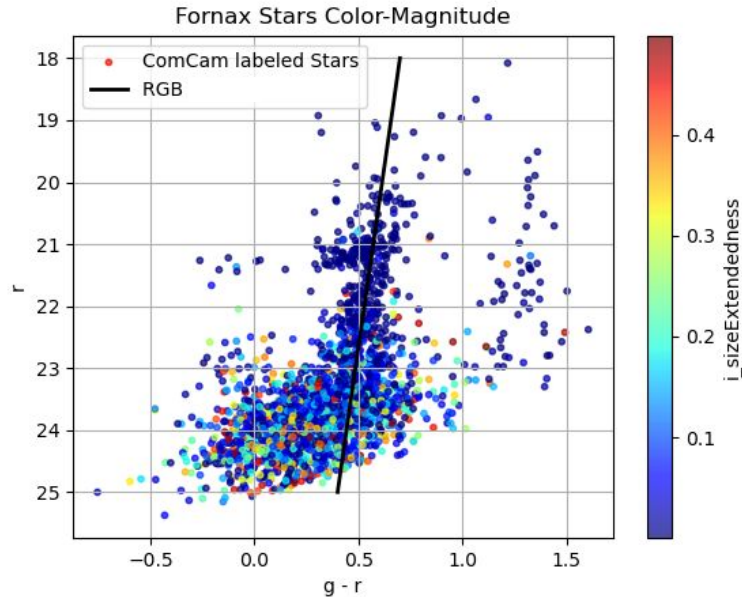
1. Queried DP1 data for objects within Fornax dSph field
 - a. Only objects where the deblend did not fail or skip, with $\text{SNR} > 5$
 - b. The hole is due to the failed deblending where there is high stellar density
2. Sorted stars and galaxies based on `i_sizeExtendedness`
 - a. For stars: < 0.5
 - b. For galaxies: > 0.5



Color-Magnitude r vs g-r

Some minor signs of contamination of objects following the RGB branch in the galaxy catalog

Also tried using $g_extendedness$ and $r_extendedness$ as the S-G classification (stars = 0, galaxies = 1), not a huge difference



Next Steps

- The next steps listed on Sprint Day were:
 - 1) visually inspecting sources, 2) using Euclid star-galaxy sep to compare (see other set of slides), 3) improving how our pipelines sort
- Analysis notebook: https://github.com/keexcell/star-gal_sep
 - Cleaning this up and documenting it
 - This analysis was done on USDF but can be recreated on NERSC or RSP (even just portal interface)